

HARSHA CHANDIMAL

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Professional Summary

Computer Science undergraduate at the University of Westminster, United Kingdom, delivered in Sri Lanka, with a strong focus on full-stack development and machine learning. Hands-on experience building real-world software systems using React, Laravel, MySQL, Node.js, Python, FastAPI, and modern AI/ML frameworks. Passionate about developing scalable, user-focused applications with clean architecture, maintainable code, and high-quality UI/UX. Experienced in building full-stack platforms, AI-powered tools, runtime code analysis systems, demand forecasting models, and machine learning prediction systems.

Education

BSc (Hons) Computer Science

2027

University of Westminster, United Kingdom — Delivered in Sri Lanka

Relevant areas: Software development, algorithms, object-oriented programming, machine learning, web development, databases, data structures, REST APIs, and software engineering.

Diploma in Information Technology

2023

Swinburne University of Technology, Australia

School

2008 – 2021

Royal College Colombo 07

Technical Skills

Languages

JavaScript, TypeScript, Python, PHP, Java, SQL

Databases

MySQL, PostgreSQL, MongoDB

Core Concepts

Full-stack architecture, API design, authentication, role-based access control, clean code, data preprocessing, model training, runtime tracing, complexity analysis, OOP, design patterns

Frontend

React.js, Vite, Tailwind CSS, React Router, Redux Toolkit, Axios

Machine Learning

Scikit-learn, PyTorch, Pandas, NumPy, RNN, LSTM, GRU, Random Forest, Logistic Regression

Backend

Laravel, Node.js, Express.js, FastAPI, REST APIs, JWT Authentication

Tools & Platforms

Git, GitHub, Docker, Railway, WAMP, Postman, VS Code, Claude Code, Antigravity

ADDITIONAL EXPERIENCE & STRENGTHS

- Strong practical experience building full-stack and AI/ML projects.
- Comfortable working with React, Laravel, MySQL, Node.js, Python, and FastAPI.
- Strong interest in clean architecture, scalable backend systems, and premium UI/UX design.
- Experience leading university group project work and managing software development tasks.
- Continuously improving skills in machine learning, backend architecture, and modern web development.
- Strong problem-solving skills with the ability to analyze technical challenges and develop practical solutions.
- Quick learner with strong adaptability to new technologies, frameworks, and development environments.

Projects

CodeMind — Code Analysis & Runtime Trace Platform

Tech Stack: React, Tailwind CSS, Laravel, MySQL, Node.js, TypeScript

Built a full-stack code analysis platform that helps users understand source code complexity, execution flow, and runtime behavior. The system supports analysis for Python, Java, and JavaScript and is designed to provide clear explanations for students and developers.

- Developed features for time and space complexity analysis, detected patterns, explanation generation, and saved analysis history.
- Built a runtime trace player to visually explain how code executes step by step.
- Designed support for variables, call stack tracking, recursion behavior, and execution steps.
- Implemented secure authentication, protected routes, analysis history management, and backend API integration.
- Focused on maintainable frontend architecture with reusable components and modern UI/UX design.
- Used clean backend structure with DTOs, services, resources, controllers, validation layers, and API resources.

DemandFlow AI — Business Demand Forecasting & Inventory Prediction System

Tech Stack: React, Laravel, MySQL, FastAPI, Python, PyTorch, Pandas, NumPy

Developed an AI/ML-based demand forecasting system designed to help businesses predict future product demand and support inventory planning.

- Built a time-series forecasting pipeline using RNN, LSTM, and GRU concepts.
- Created data preprocessing workflows including date parsing, sorting, feature engineering, scaling, and sequence creation.
- Integrated a Python/FastAPI machine learning service with a full-stack web application.
- Designed the system to support future migration from synthetic business data to real-world retail datasets.
- Focused on understanding and implementing the complete machine learning pipeline from data preparation to model prediction.

FIFA Match Predictor — Machine Learning Football Match Outcome Prediction System

Tech Stack: Python, Scikit-learn, FastAPI, React, Machine Learning

Built a machine learning system to predict football match outcomes using historical match data and ensemble modeling techniques.

- Developed a hybrid ensemble model combining Random Forest and Logistic Regression probabilities.
- Achieved a weighted F1 score of 0.5589 and macro F1 score of 0.5223.
- Improved draw detection with draw recall of 42.49%.
- Integrated the trained model into a FastAPI backend for production prediction usage.
- Designed the user-facing system around Team A vs Team B logic instead of home/away terminology for better FIFA match representation.

AutoFix / Vehicle Ecosystem Platform — Full-Stack Vehicle Service Marketplace

Tech Stack: React, Tailwind CSS, Node.js, Express.js, Laravel, MySQL, PostgreSQL, JWT

Built a full-stack vehicle service platform connecting vehicle owners with garages, mechanics, detailers, and vehicle carriers.

- Developed features for provider registration, vehicle management, service listings, bookings, service status tracking, and invoice generation.
- Implemented authentication and role-based access control for owners and service providers.
- Created booking workflows with service date, time slot, status transitions, and provider actions.
- Built owner and provider dashboards with service history and booking management.
- Designed reusable frontend components for provider cards, filters, booking pages, invoice modals, and status badges.

CAREER INTERESTS

Full-Stack Developer | Software Engineer | Machine Learning Engineer | AI/ML Developer | Backend Developer